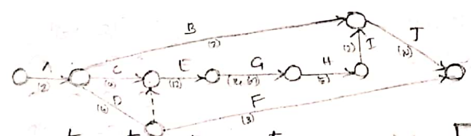
Q)

Activity	Predecessor
A	–
B	A
C	A
D	A
E	C, D
F	D
G	E
H	G
I	H
J	B, I



	t_o	t_m	t_p	t_e
A	1	2	3	2
B	1	2	3	2
C	2	4	6	4
D	2	5	14	6
E	6	12	18	12
F	1	3	5	3
G	10	12	30	14.67
H	3	5	7	5
I	1	2	3	2
J	5	10	15	10

$$t_e = \frac{t_o + 4t_m + t_p}{6}$$

Scanned by CamScanner

6) Activity — Duration

Activity	Duration
1–2	2
1–4	2
1–7	1
2–3	4
3–5	1
4–6	5
4–8	8
5–6	4
6–9	3
7–8	3
8–9	3

critical path

1–2–3–5–6–9 = 14

1–4–6–9 = 10

1–4–8–9 = 13

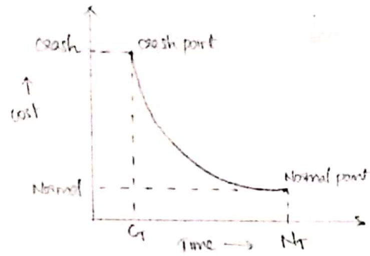
1–7–8–9 = 7

Float = LST – EST

7)

Activity	Normal Time	Normal cost	Crash Time	Crash cost	cost slope
1 – 2	8	100	6	200	50
1 – 3	4	150	2	350	100
2 – 4	2	50	1	70	40
2 – 5	10	100	5	400	60
3 – 4	5	100	1	200	25
4 – 5	3	80	1	100	10
		580			

$$\text{cost slope} = \frac{\text{cost diff}}{\text{Time diff}}$$



7) Let the indirect cost is 70/day

Total normal cost = 580

Total duration (critical path) = 18

Total Indirect cost = 70×18

= ~~1840~~ 1260

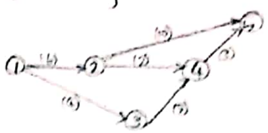
Total cost = 1840

Crashing

check cost slope of critical activities 1-2 & 2-5 both are less than 70

First take lowest cost slope activity 1-2 & crash

Crashing activity 1-2 to 8→6



DC = 580 + 2×50 → extra cost for crashing of 2 days

= 680

IC = 70 × 16 → new critical path

= 1120

TC = 1800

2-5 10→5

DC = 580 + 100 + 5×60 + 2×10

= 980 + 20 = 1000

IC = 70 × 11

= ~~782~~ 770

TC = ~~[illegible]~~ 1770

7)

Activity	Normal		Crash		Cost slope
	Time	Cost	Time	Cost	
1 - 2	8	300	2	[illegible]	100
2 - 3	5	30	5	30	0
2 - 4	7	420	5	580	80
2 - 5	9	720	7	810	45
3 - 4	5	250	4	300	50
4 - 5	0	0	0	0	0
4 - 6	6	570	4	660	45
5 - 7	9	[illegible]	7	470	70
6 - 8	13	780	10	900	40
7 - 8	10	1000	9	1200	200
		4150			

The following data gives details about normal and crash cost and time of a project. Draw the network and identify the critical path. What is the normal project duration and cost. Crash the relevant activities and determine the optimal project duration and cost. Assume the indirect cost as 100

critical path.

By crashing 5-6 from 6 to 4

critical path 1-2-5-6-7-8 (30)

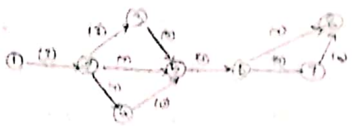
DC = 4220 + 2×45

= 4310

ID = 150×30 = 4500

TC = 8810

crashing 2-4 from 9 to 7



critical path 1-2-3-5-6-7-8

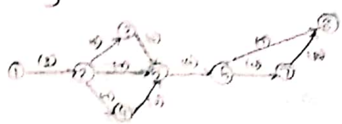
DC = 4220 + 2×45 + 2×45

= 4400

ID = 150×29 = 4350

TC = 8750

crashing 3-5 from 5 to 4



critical path 1-2-3-5-6-7-8

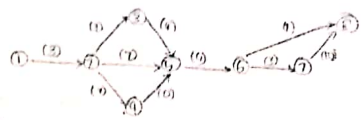
DC = ~~420~~ 4220 + 2×45 + 2×45 + 1×50

= 4450

IC = 150×28 = 4200

TC = 8650

crashing 6-7 from 4 to 3



critical path 1-2-3-5-6-7-8

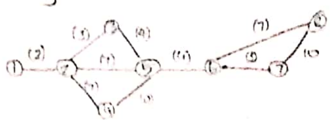
DC = 4450 + 1×70

= 4520

IC = 150×27 = 4050

TC = 8100

crashing 1-2 from 3 to 2



Critical path 1-2-3-5-6-7-8

DC = 4520 + 1x 100

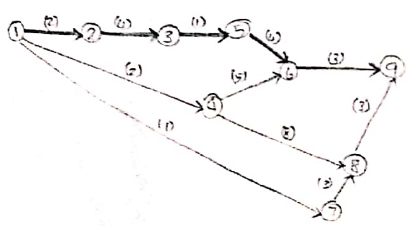
= $\underline{4620}$

IC = 150 × 26 = $\underline{3900}$

TC = $\underline{8520}$

?)

Activity	Duration	ES	LS	Slack (LS-ES) / Float (EF-LF)	EF	LF
1-2	2	0	0		2	2
1-4	2	0	4		2	3
1-7	1	0	7		1	8
2-3	4	2	2		6	6
3-5	1	6	6		7	7
4-6	5	2	6		7	11
4-8	8	2	3		10	11
5-6	4	7	7		11	11
6-9	3	11	11		14	14
7-8	3	1	8		4	11
8-9	3	4	11		7	14



Critical path
1-2-3-5-6-7 (14)

Forecasting

Judgemental Techniques — Long-term

Time series models — Short-term

Causal method

Judgemental tech

- Opinion survey
- Customer survey
- Distributors
- Marketing Trials
- Market research
- Delphi method

- Least square method
- Moving average
- Weighted moving avg
- Exponential smoothing
- Correlation

?) The following data gives the sales of a company for various years. Fit the straight line and forecast the sales for the year 2018 & 19.

year	sales (in lakhs)	x	x^2	xy
2009	13	-4	16	-52
10	20	-3	9	-60
11	20	-2	4	-40
12	28	-1	1	-28
13	30	0	0	0
14	32	1	1	32
15	33	2	4	66
16	38	3	9	114
17	43	4	16	172
	257		60	204

$\boxed{y = a + bx}$

$a = \frac{\Sigma Y}{N}$

$b = \frac{\Sigma xY}{\Sigma x^2}$

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$a = \frac{257}{9} = \underline{28.5}$

$b = \frac{324}{60} = \underline{5.4}$

$y = \underline{28.5 + 5.4x}$

For 2018, $x = 5$

$y = 28.5 + 5.4 \times 5$

$= \underline{45.5}$

For 2019, $x = 6$

$y = 28.5 + 5.4 \times 6$

$= \underline{48.9}$

<u>Moving avg method</u>

Year	Sales	Moving avg
2001	580	
2002	585	591
2003	610	623
2004	675	678
2005	750	762
2006	860	860
2007	970	

From the given data

a) compute 3 months year moving avg and find out the forcasted sales for year 2008 & 2009

b) Also find out 3 year weighted moving avg for the year 2008. where the weights are 0.5 for the latest year. 0.3 & 0.2 for other years respectively.

2018

Months	Jan	Feb	Mar	Apr	May	Jun	July	Aug	Sep	Oct	Nov	Dec
Sales	400	490	570	500	640	680	710	800	820	910	860	950
$y \rightarrow$		486	520	570	606	676	730	776	843	863	906	
$x \rightarrow$		-4.5	-3.5	-2.5	-1.5	-0.5	0.5	1.5	2.5	3.5	4.5	

Data given in the table represents sales figure of a company for 12 months of year 2018

i) compute 5 months moving avg (ignore decimal)

ii) Forecast the demand of the month of Jan 2019

3) Actual demand for the month of Jan 2019 905 units what should be the forecast for the month of Feb 2019

$a = \frac{\Sigma y}{N}$

$b = \frac{\Sigma xy}{\Sigma x^2}$

Exponential smoothing

Forecast for the period t (F_t) =

Forecasted demand past period $(F_{t-1}) + \alpha(D_{t-1} - F_{t-1})$

α smoothing constant

0.01 - 0.1

0.1 - 0.7

?) The demand for a particular product is 300 units & 350 units for the months of sept & oct respectively. Using 200 units as forecast for sept compute forecast for the month of Nov assume the value of smoothing constant as 0.7

Forecast of October = 200 + 0.7 × (300 - 200)

= 270

Forecast for November = 270 + 0.7 × (350 - 270)

= 326

?) The following data relates the cost of productn & sales prices

Productn cost	
203	225
216	242
223	250
239	271
248	275
253	277
279	295
301	318

x Prodn cost	203	216	223	239	248	253	279	301	311
y sales price	225	242	250	271	275	277	295	318	329
$x - \bar{x}$	-49.55	-36.55	-29.55	-13.55	-4.55	0.45	26.45	48.45	58.45
$y - \bar{y}$	-46.33	-29.33	-21.33	-0.33	3.67	5.67	-16.33	46.67	57.67

$$r = \frac{\sum xy}{\sqrt{\sum x^2 \cdot \sum y^2}}$$

$$r = \frac{\sum XY -}{\sqrt{\sum X^2 - \sum Y^2}}$$

$X = x - \bar{x}$

$Y = y - \bar{y}$

$\bar{x} = 292.55$

$\bar{y} = 271.33$

Maintenance Management

Maintenance is classified into

1) Breakdown maintenance
2) Preventive "
3) Predictable "
4) opportunity "
5) shutdown "

Replacement

- operating cost more
- continuous failure
- Technology changing
- Efficiency low

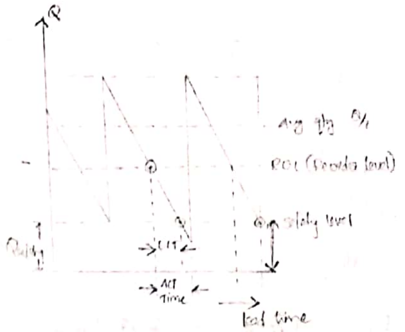
Stages in product design

Conception ← Idea generation, draft specification
↓
Acceptance ← Evaluation & screening
↓
Execution ← model
↓
Translation ← Prodⁿ dept machine
↓
Pre-prodⁿ ← Pilot run

Factors influencing product design

Marketing aspects → Product design
Function of aspects or Product characteristics → Product design
Economical aspects → Product design
Aesthetic aspects → Product design

Inventory control



Costs

Cost of production, C_p = Rs/unit

Ordering cost, C_o = Rs/order

Holding cost, C_h = Rs/unit/annum

Annual demand (D)

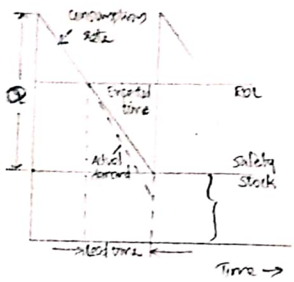
Total cost = Ordering cost + Holding cost

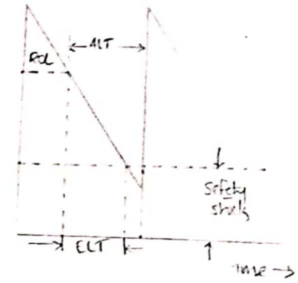
$$\boxed{T_c = C_o \cdot \frac{D}{Q} + \frac{Q}{2} \cdot C_h}$$

Demand

$$\frac{d(T_c)}{dQ} = C_o \cdot \frac{-D}{Q^2} + \frac{C_h}{2} = 0$$

$$\boxed{Q = \sqrt{\frac{2 C_o D}{C_h}}}$$ economic order quantity





?) A manufacturer has to supply 3600 units / year shortages are not permitted inventory carrying cost amount ₹ 1.2 per unit per annum. The ~~second~~ setup cost per run is ₹ 80 (ordering cost). Find

i) EOQ
ii) optimum no. of orders per annum
iii) Min avg ~~cost~~ annual inventory cost
iv) opt period of supply / opt order

$$Q^* = \sqrt{\frac{2C_oD}{C_h}}$$

$$= \sqrt{\frac{2\times 80\times 3600}{1.2}} = \cancel{828.63}\ \underline{692.82}$$

$$\text{Cost} = C_o\frac{D}{Q^*} + \frac{Q^*}{2}C_h$$

$$= 80\times\frac{3600}{692.82} + \frac{692.82}{2}\times 1.2 = \underline{\underline{831.38}}$$

optimum no of orders $= \frac{D}{Q^*}$

$$= \frac{3600}{692.82} = 5.19 \approx 5 \text{ orders}$$

$$T_{cm} = \sqrt{2DC_oC_h}$$

$$= \sqrt{2\times 3600\times 80\times 1.2} = \underline{\underline{831.38}}$$

opt period of supply per opt order

$$\frac{1}{N} = 0.2 \text{ years.}$$

?) Annual demand of product is 10000 per year (D)
cost per unit is ₹ 2 (C_p)
ordering cost is ₹ 36 (C_o)
Inventory carrying cost (holding cost) is estimated at 9 % of avg inventory investment $\quad C_h = C_pI \quad (I = 9\%)$
opt no of orders per annum, Min total cost of inventory per annum

$$C_h = C_pI$$

$$= 2\times 0.09 = 0.18$$

$$Q = \sqrt{\frac{2C_oD}{C_h}}$$

$$= \sqrt{\frac{2\times 36\times 10000}{0.18}} = \underline{\underline{2000}}$$

$$T = \sqrt{2DC_0C_h}$$

$$= \sqrt{2 \times 10000 \times 36 \times .18} = \underline{360}$$

opt no of orders $= \dfrac{D}{Q^*}$

$$= \frac{10000}{2000} = \underline{5}$$

opt period of supply per opt order

$$\frac{1}{N} = \underline{0.2 \text{ years}}$$

2) A company needs 600 units per month, procurement (ordering) cost is ₹ 36 per order. cost of holding it in stock is ₹ 1.2 per unit per year.
Determine the opt quantity
If the consumption of above item increases to 40 no. per day and the actual inventory holding cost is 0.50 per unit per month. what will be the revised EOQ

$D = 600$

$C_0 = 36$

$C_h = 1.2$

$$Q^* = \sqrt{\frac{2C_0D}{C_h}} = \sqrt{\frac{2 \times 36 \times 600}{1.2}} = \underline{189.73}$$

$D' =$ ~~40×365 = 1460 per year~~ $40 \times 365 = \underline{14600}$

$C_h' = 0.5 \times 12 = \underline{6}$

$$Q^* = \sqrt{\frac{2C_0D}{C_h}} = \sqrt{\frac{2 \times 36 \times 14600}{6}} = \underline{418.56}$$

<u>Seminar</u>

Explain about
sloc

Sales programme
Master schedule
Resource requirement plan
↓
Loading
↓
Sequencing
↓
Scheduling
↓
Expediting

<u>n jobs & 2 machines</u>

Job	M_1	M_2
~~A~~	5	2
~~B~~	1	6
~~C~~	9	7
~~D~~	3	8
~~E~~	10	4

Johnson's Algorithm

B	D	C	E	A

Job	M_1		M_2		Idle time
	in	out	in	out	
B	0	1	1	7	1
D	1	9	7	15	0
C	9	13	15	22	0
E	13	23	23	27	1
A	23	28	28	30	1

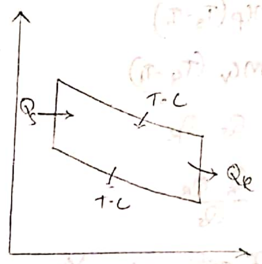
% (2)

Job	m_1	m_2
A	6	16
B	24	20
C	30	20
D	12	13
E	20	24
F	22	2
G	18	6

A	D	E	C	B	G	F

A

Job	M_1		M_2		idle time.
	in	out	in	out	
A	0	6	6	12	
B D	6	18			
C E	18	38			
D C	38	61			
E B	68	92			
F G	92	110			
G F	110	152			



Engine performance

Break power (kW) $bp = \frac{2\pi NT}{60000}$, $T = (W-S)R$

Indicated power (kW) $ip = \frac{L A n k P_{im}}{60 \times 1000}$

Indicated thermal efficiency, $\eta_{ith} = \frac{ip}{\text{energy - input/s}}$

$= \frac{ip}{m_f \times C.V}$

Break thermal efficiency, $\eta_{bth} = \frac{bp}{m_f \times C.V}$

Mech efficiency, $\eta_m = \frac{bp}{bp + fp}$

Volumetric efficiency,

$\eta_v = \frac{\text{mass of charge actually inducted}}{\text{mass of charge in swept volume at ambient}}$

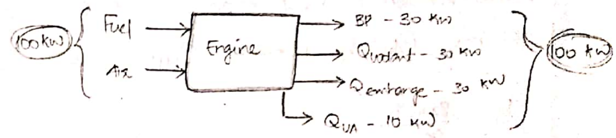
Relative efficiency, $\eta_{rel} = \frac{\text{Actual thermal efficiency}}{\text{Air standard efficiency}}$

mean effective pressure, $P_{im} = \frac{60 \times 1000 \times ip}{L A n k}$

Specific fuel consumption SFC = $\frac{\text{fuel consumption per unit time}}{\text{power}}$

Air-fuel ratio = $\frac{\text{mass of air}}{\text{mass of fuel}}$

Heat balance



Heat input = $m_f \times C.V$

$Q_{coolant} = m_w C_{pw} (t_o - t_i)$

$Q_{exhaust} = m_g C_{pg} (T_g - T_a)$

Frictional Power

Morse Test

- Multiple cylinder engine
- For SI & CI engine
- Isolate one cylinder, adjust load to bring rated speed.

$bp_k = \sum_1^k ip_k - fp$

$bp_{k-1} = \sum_2^k ip_k - fp$

$ip_1 = bp_k - bp_{k-1}$

$ip_k = \sum_1^k ip_k$

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?) The air flow to a 4 cylinder 4 stroke oil engine is measured with an orifice dia 5 cm and Cd 0.6. The engine bore is 10 cm, stroke 12 cm, speed 1200 rpm Brake torque 120 Nm, fuel consumption 5 kg/h CV of fuel 42 MJ/kg. Pressure drop across orifice is 4.6 cm of water ambient is at 17° C and 1 bar. Calculate brake thermal efficiency, break mean effective pressure, volumetric efficiency (An: bp = 15.08 kW, η_{bth} = 25.85%, P_{m} = 4 bar η_v = 85.7%)

$$bp = \frac{2\pi NT}{60}$$

$$= \frac{2\pi \times 1200 \times 120}{60} = 15079.64 = 15.08 \text{ kW}$$

$$\eta_{bth} = \frac{bp}{m_f \times CV}$$

$$m_f = 5 \text{ kg/h}$$

$$= 5/3600 = 1.38 \times 10^{-3} \text{ kg/s}$$

$$= \frac{15.08 \times 10^3}{42 \times 10^6 \times 1.38 \times 10^{-3}} = 25.84\%$$

$$\cancel{\frac{ip = LAnKP_m}{60}}$$

$$P_m = \frac{60 \times ip}{LAnK}$$

$$= \frac{60 \times 15.08}{}$$

?) A gasoline engine working on 4 stroke develops brake power of 20.9 kW. A morse test was conducted on the engine and bp obtained when each cylinder was made inoperative by short circuiting the spark plug are 14.9, 14.3, 14.8 and 14.5 respectively. The test was conducted at constant speed. Find the indicated power, mechanical efficiency and bmep when all the cylinders are firing. Bore of the engine 75 mm, stroke 90 mm and speed 3000 rpm (An IP = 25.1 kW)

bp = 20.9 kW

ip of 1st cylinder = 20.9 − 14.9 = 6

" IInd " = 20.9 − 14.3 = 6.6

" IIIrd " = 20.9 − 14.8 = 6.1

" IVth " = 20.9 − 14.5 = 6.4

25.1 kW

$$\eta_{mech} = \frac{bp}{bp + fp}$$

$$= \frac{\cancel{20.9}}{\cancel{20.9 + 25.1}} = \frac{20.9}{25.1} \times 100 = 83.27\%$$

$$BP = \frac{LANKP}{60}$$

$$20.9 = \frac{90 \times \pi \times \frac{75^2}{4} \times \frac{3000}{2} \times 4 \times P}{60}$$

$$P = 5.25 \text{ bar}$$

?) A 4 stroke engine has a cylinder dia 25 cm and stroke 45 cm. The effective dia of the brake drum is is 1.6 m. The observations are listed down these

Duration of test	= 40 min
Total revolution	= 8080
Total explosion	= 3230
Net load on brake	= 90 kg
Mean effective pressure	= 5.8 bar
Volume of gas used	= 7.5 m^3
Pressure of gas indicated in meter	= 136 mm of water
Atm temp	= 17°C
CV of gas	= 19 MJ/m^3 at NTP
Temp rise of cooled water	= 45°C
Cooling water supplied	= 180 kg

Find the heat balance for the engine indicated thermal efficiency, brake thermal efficiency. Assume atm pressure as 760 mmHg (Ans : bp 14.94 kW, Heat supplied 3396.25 kJ/min, Heat BP = 896.4 kJ/min, Heat cooling = 846.5 kJ/min, unaccounted loss = 1653.35 kJ/min, η_{th} = 26.39%, η_{bt} = 30.49%)

$$Rev^n/min = \frac{8080}{40} = 202$$

$$\frac{8080}{2} = 4040$$

$$Rev^n/min = \frac{4040}{40} = 101$$

$$\boxed{T = 90 \times 9.8 \times 0.8 =}$$

$$\cancel{BP = \frac{2\pi NT}{60} = \frac{2\pi \times 101 \times}{60}}$$

~~T = (W-S) Re = W×R~~
~~= 90 × 0.8~~

$$BP = \frac{2\pi NT}{60}$$

$$= \frac{2\pi \times 8080 \times 90 \times 9.8 \times 0.8}{40 \times 60} = 14.92\ kW$$

We have to convert actual volume 7.5 m^3 into NTP condition and then multiply with

$$\left\{ \frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2} \right\}$$

$$\cancel{\frac{1 \times V_1}{20} = \frac{1332.8 \times 10^{-5} \times 7.5}{17}}$$

$$\cancel{V_1 = \frac{1332.8 \times 10^{-5} \times 7.5 \times 20}{17}}$$

$$\cancel{\frac{1 \times V_1}{20} = \frac{1.0265 \times 7.5}{17}}$$

$$\cancel{V_1 =}$$

~~136 mm of water~~

$$\cancel{P = h\rho g = 136 \times 10^{-3} \times 1000 \times 9.8 = 1332.8\ Pa}$$

$$\cancel{= 1332.8 \times 10^{-5}\ bar}$$

$$\cancel{P = h\rho g + 760\ mm\ Hg = 1332.8 + 101325.024 = 102657.824\ Pa = 1.0265\ bar}$$

136 mm of water = 10 mm of Hg

$P_2 = (10 \text{ mm} + 760 \text{ mm})$ of Hg

$= 770$ mm of Hg

$= 1.0265$

$Q_{in} = CV \times V_2$

$= \frac{19 \times 7.67}{40} = 3.6432$ MJ/min

$= 3643.2$ KJ/min

$Q_{BP} = BP \times 60$

$= 895.2$ KJ/min

$Q_{w} = m C_p W (\Delta T)$ $[C_p = 4.184$ KJ/kg°C$]$

$= \cancel{180 \times 4.184 (\ldots)}$

$= \cancel{44.92\ \text{kW}}\ \frac{180}{4} \times 4.184 \times 45$

$= \cancel{8472.6 \text{ KJ/min}}$ 847.26 KJ/min

$Q_{unaccountable} = Q_{in} - \text{Exhaust}$

$= 3643.2 - 1744.2$

$= 1899$ KJ/min

2) The observations of a retardation test on a single cylinder diesel engine are as follows, Rated power 8 kW. Calculate frictional power and mechanical efficiency.

Sl. No	Drop in speed (rpm)	Time for fall - no load (s)	Time for fall - 50% load (s)
1	475 → 400	7	2.2
2	475 → 350	10.6	3.7
3	475 → 325	12.5	4.8
4	475 → 300	15	5.4
5	475 → 275	16.6	6.5
6	475 → 250	18.9	7.2

$T_f = \frac{t_3}{t_2 - t_3} \times T_{50\%}$

$= \frac{2.2}{7 - 2.2} \times$